CLASS 10 · CBSE · NCERT MATHEMATICS · FORMULA BANK

Chapter 13: Statistics

Every formula and result of the chapter on one sheet

1	Class mark	$x_i = \frac{\text{lower limit} + \text{upper limit}}{2}$
2	Class size	$h = \text{upper limit} - \text{lower limit}$
3	Mean: direct method	$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$
4	Mean: assumed mean method	$\bar{x} = a + \frac{\sum f_i d_i}{\sum f_i}$, where $d_i = x_i - a$
5	Mean: step-deviation method	$\bar{x} = a + \left(\frac{\sum f_i u_i}{\sum f_i}\right) \times h$, where $u_i = \frac{x_i - a}{h}$
6	Mode	$l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$
7	Median	$l + \left(\frac{\frac{n}{2} - cf}{f}\right) \times h$
8	Modal class	The class with the highest frequency
9	Median class	The class whose cumulative frequency is just greater than $\frac{n}{2}$
10	Empirical relationship	$3\,\text{Median} = \text{Mode} + 2\,\text{Mean}$
11	Making classes continuous	Subtract 0.5 from each lower limit and add 0.5 to each upper limit (for a gap of 1)